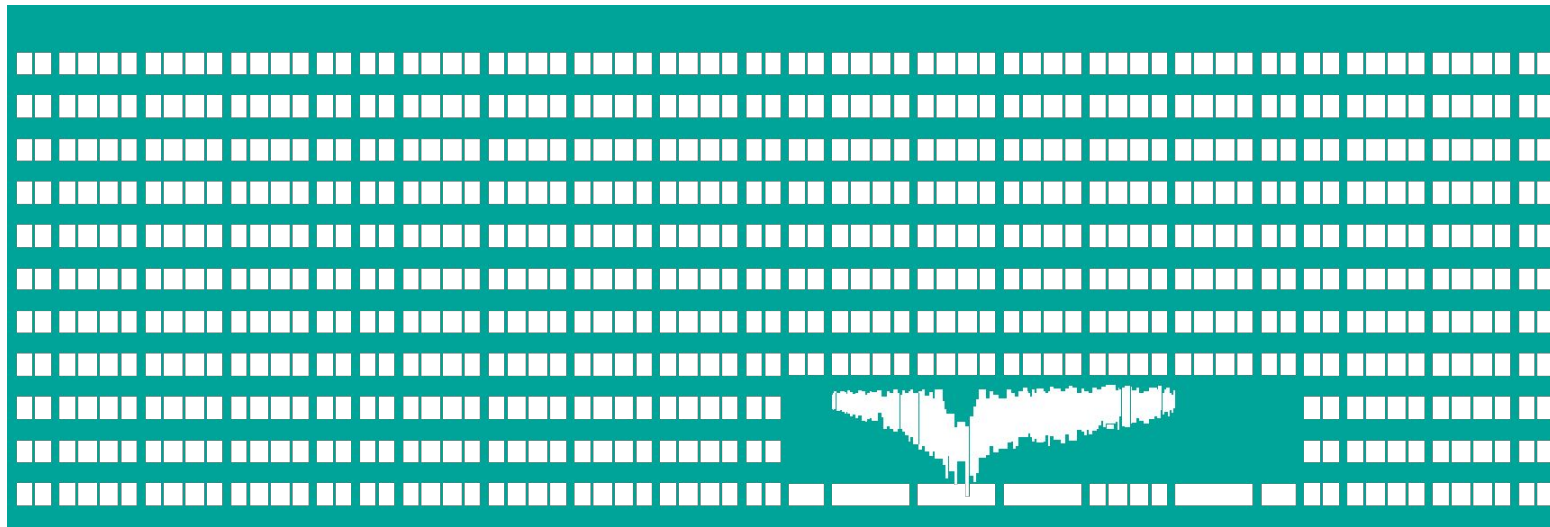




05 - Speech Quality Measurement

prof. Ing. Miroslav Vozňák, Ph.D.

Classification of methods

- generally subjective and objective methods
- MOS scale ITU-T P.800,
- ve grades (1-5) : excellent, very good, good, low and very low speech quality
- nal MOS is calculated as a mean of all MOS values



Classification of methods

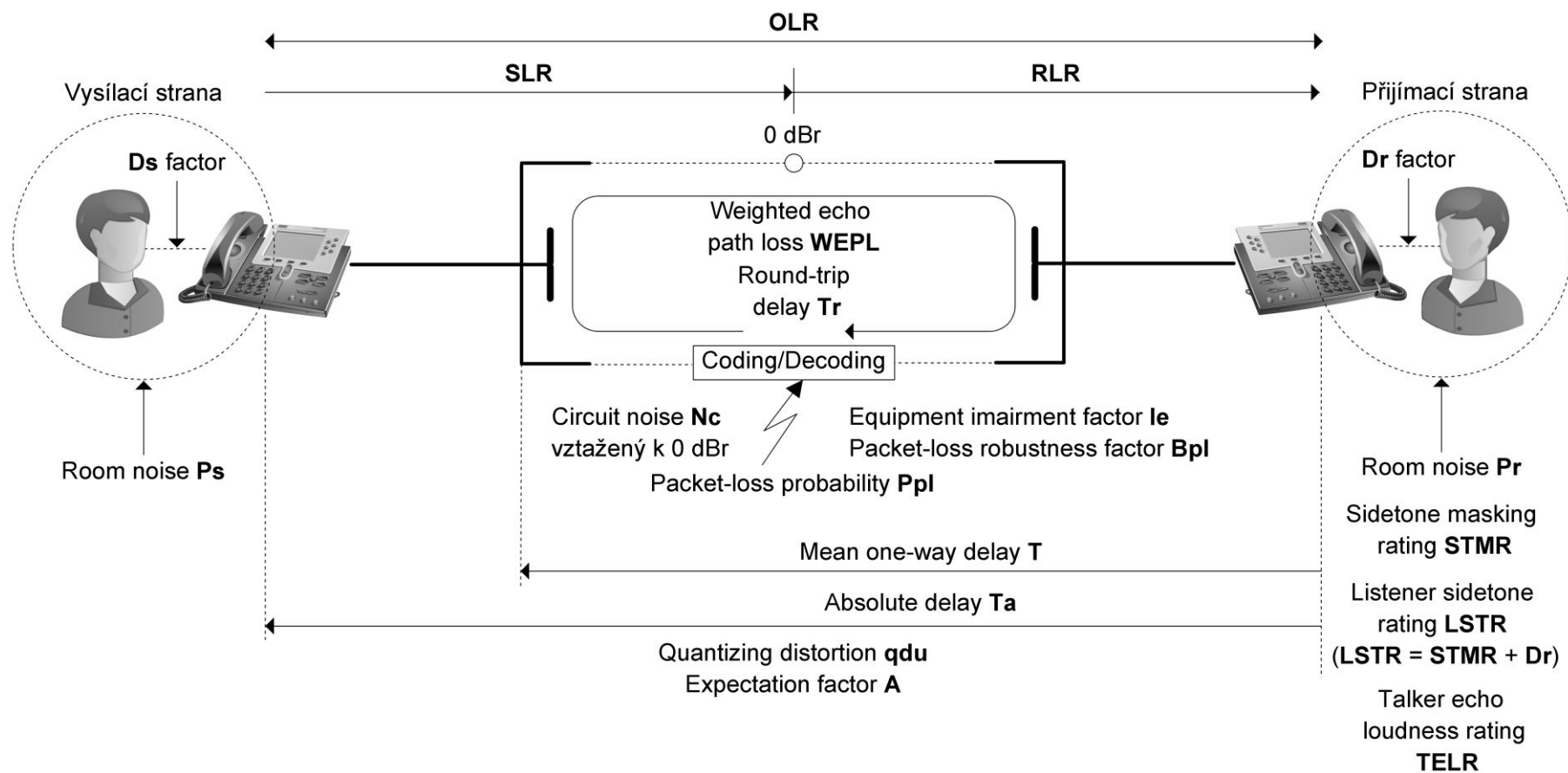
- Listening or Conversational Quality,
- Objective, Subjective or Estimated
- MOS-LQO Listening Quality Objective
- MOS-CQE Conversational Quality Estimated

Methods of Assessment	Listening	Conversational
Subjective	MOS-LQS	MOS-CQS
Objective	MOS-LQO	MOS-CQO
Estimated	MOS-LQE	MOS-CQE

Approach

- Subjective:
 - ACR (Absolute Category Rating)
 - DCR (Degradation Category Rating)
- Objective:
 - Intrusive (PESQ ITU-T P.862, POLQA P.863)
 - Non-intrusive (E-Model ITU-T G.107)

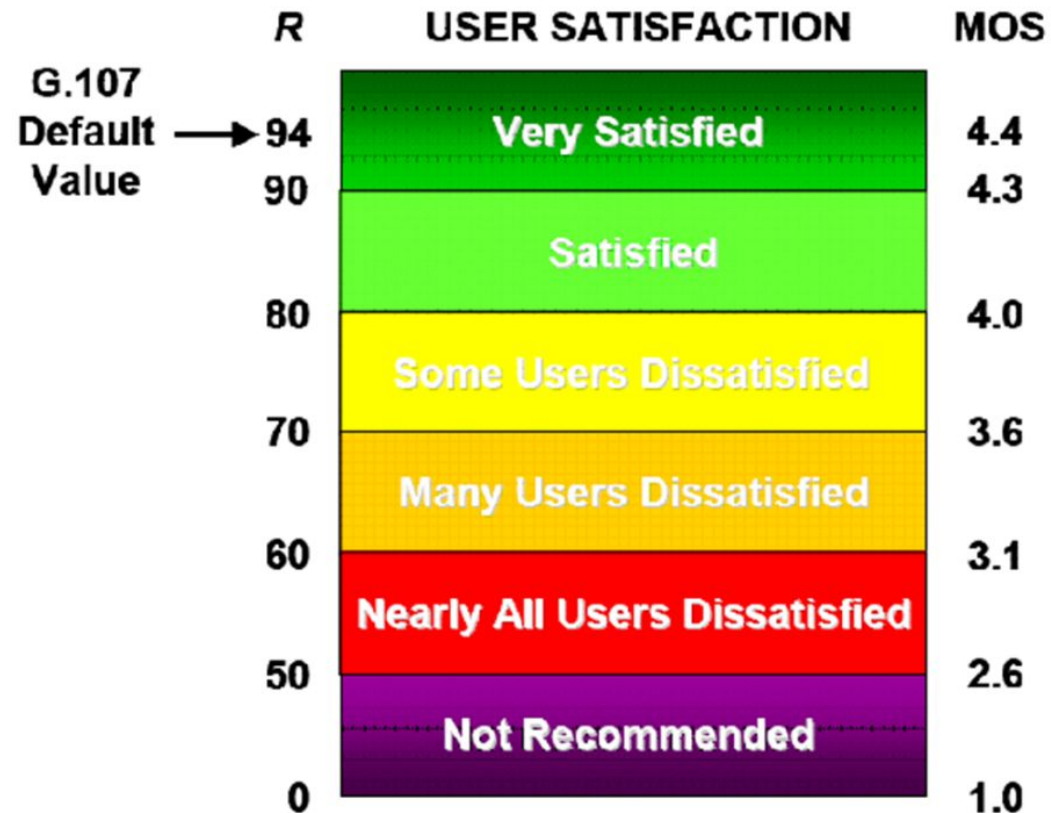
E-Model



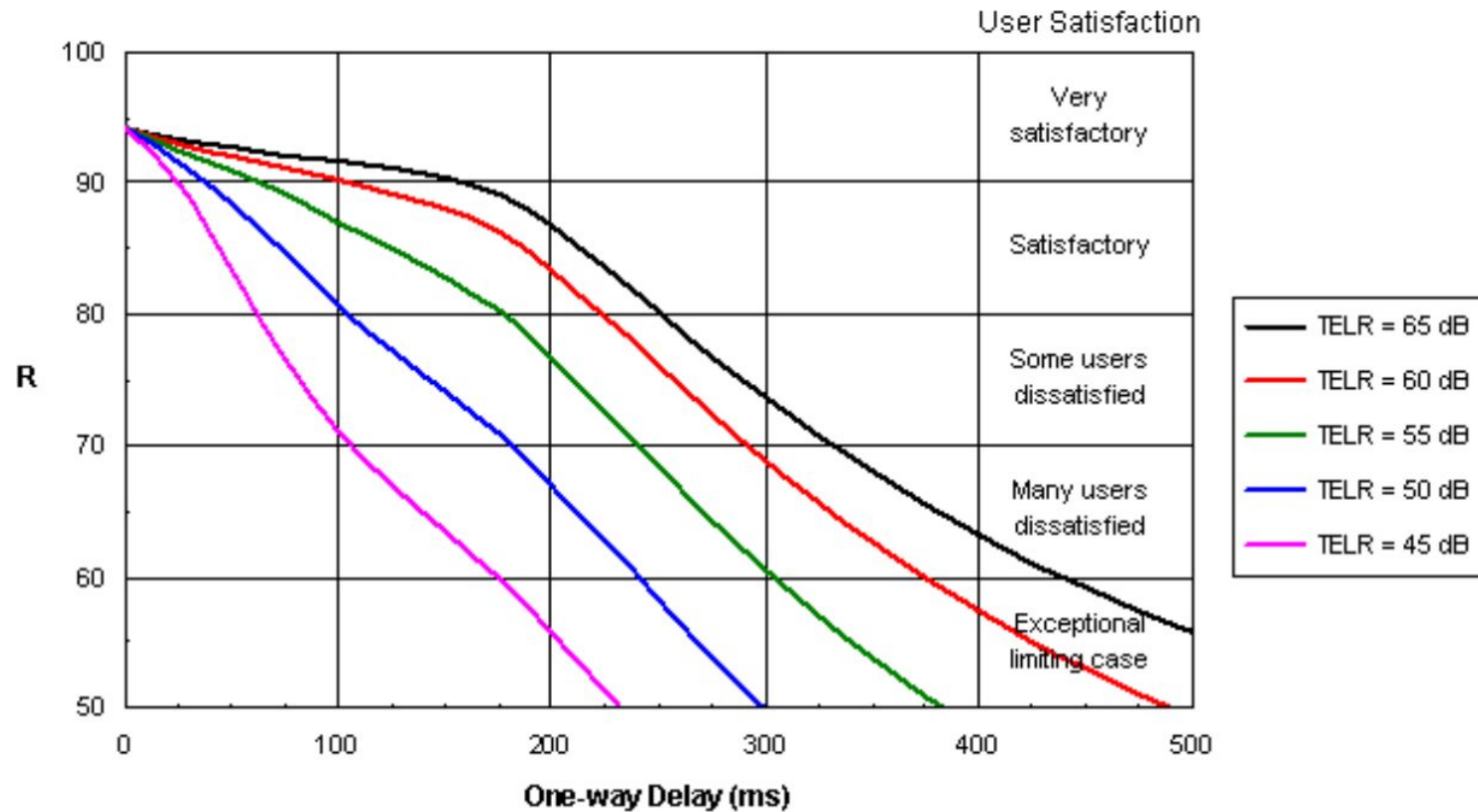
E-Model

- SLR [dB] (Send Loudness Rating), RLR [dB] (Receive Loudness Rating)
- OLR [dB] (Overall Loudness Rating) $OLR = SLR + RLR$
- T [ms] (Mean one-way Delay) , Ta [ms](Absolute Delay)
- TELR [dB] (Talker Echo Loudness Rating),
- Ie [-] (Equipment Impairment Factor)
- Bpl [-] (Packet-loss Robustness Factor), A [-] (Expectation factor)
- Ppl [%] (Packet-loss Probability), BurstR [-] (Burst Ratio)

E-Model and R-factor vs MOS



E-Model and R-factor vs delay



E-Model Calculation according to G.107

$$R = R_o - I_s - I_d - I_{e-eff} + A$$

- R, R-factor, result 0-100
- R_o , SNR
- Impairments I_s (Simultaneously), I_d (caused by Delay)
- I_{e-e} (Efficient Equipment Impairment)
- advantage factor A
- I_{e-e} impairment is defined as:

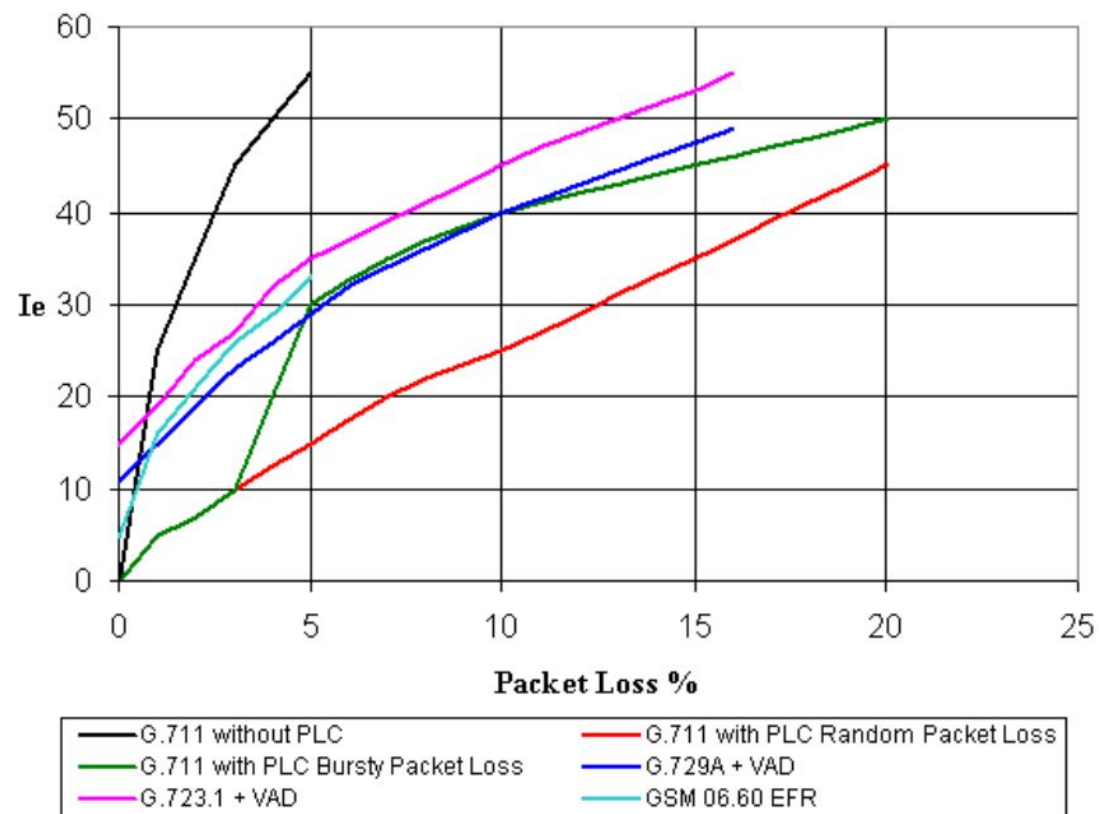
$$I_{e-eff} = I_e + (95 - I_e) \cdot \frac{P_{pl}}{\frac{P_{pl}}{BurstR} + B_{pl}}$$

- I_e is Impairment factor (tabelized in G.113), P_p represents the packet loss, B_{pl} is the Packet-LossRobustness factor (again in G.113) and $BurstR = 1$ if the I loss is random (for burstiness $BurstR > 1$).

le - Equipment Impairment

CODEC	Bit rate [kbps]	le
ADPCM (G.726, G.727)	40	2
ADPCM (G.721, G.726, G.727)	32	7
ADPCM (G.726, G.727)	24	25
ADPCM (G.726, G.727)	16	50
LD-CELP (G.728)	16	7
LD-CELP (G.728)	12.8	20
CS-ACELP (G.729)	8	10
CS-ACELP (G.729A + VAD)	8	11
ACELP (G.723.1)	5.3	19
MP-MLQ (G.723.1)	6.3	15

Ie vs Packet loss



B_{pl} - Packet Loss Robustness and A-factor

CODEC	le	t [ms]	Bpl]
G.723.1 MP-MLQ	15	30	16.1
G.729 CS-ACELP	10	20	19
GSM EFR	5	20	10
G.711 without PLC	0	10	4.3
G.711 with PLC	0	10	25.1

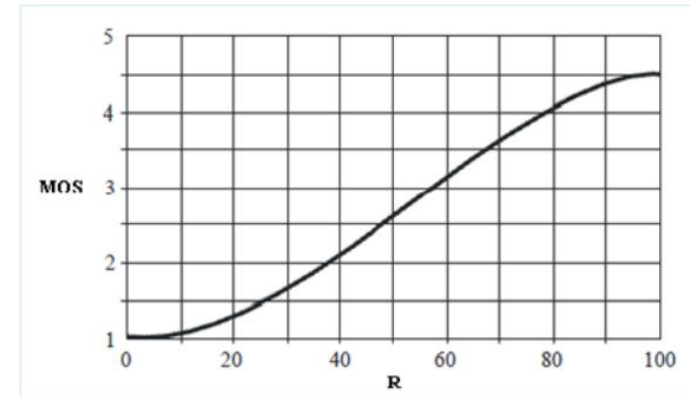
Equipment	A
Fixed terminal	15
Mobile terminal in building (DECT)	5
Mobile terminal outside or moving (Cell Phone)	15
Satellite phones	20

Simplified E-Model

- $R_0 = 94.7688$ by default
- $I_s = 1.4136$ by default
- I_d negligible if delay is less than 100 ms, $I_d = 0$
- BurstR = 1 (ideal loss distribution)

$$MOS = \begin{cases} 1 + 0,035 \cdot R + R \cdot (R - 60) \cdot (100 - R) \cdot 7 \cdot 10^{-6} & R < 100 \\ 4,5 & R > 100 \end{cases}$$

Transform between R-factor and
MOS-CQE



Thank you for your attention

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